

BT503 Current Papers Spring 2020

1. Define Halophiles.

2

Halophiles... survive intensely saline environments, such as exist in natural salt lakes or salt evaporation ponds.

2. Define psychrophiles.

2

Psychrophiles can tolerate these circumstances.

3. Define Biosensors.

2

A biosensor is an analytical device, used for the detection of a chemical substance, that combines a biological component with a physicochemical detector.

4. Define co-metabolism.

2

Co-metabolism is defined as the simultaneous degradation of two compounds, in which the degradation of the second compound (the secondary substrate) depends on the presence of the first compound (the primary substrate).

5. How microbes are poly cyclic aromatic compounds.

2

Various bacteria have been found to degrade PAHs, in which degradation of naphthalene and phenanthrene has been most widely studied. ... Principally, bacteria favor aerobic conditions for degradation of PAHs via oxygenasemediated metabolism

6. Write two main chemicals characteristics of chlorinated pesticides that lead to problems. 2

Two characteristics of chlorinated pesticides that led to great harm are *their resistance to biodegradation and their hydrophobicity*

7. Write two advantages of the conventional biological sludge dewatering process. 2

8. What is meant by minimum substrate concentration? 2

9. Write down the ways to achieve goal of environment biotechnology. 2

10. Write the main characteristics of intensive technology. 2

Fast response and low treatment time

11. How plants are modified as bioindicator. 2

Plant indicators The presence or absence of certain **plant** or other vegetative life in an ecosystem can provide important clues about the health of the environment: environmental preservation. There are several types of **plant** biomonitors, including mosses, lichens, tree bark, bark pockets, tree rings, and leaves.

12. Mention two Natural sludge dewatering processes. 2

Natural sludge dewatering processes:

- *Drying beds*
- *Sludge lagoons*

13. Write two sources of dioxins? 2

- waste-burning incinerators of various sorts
- backyard burn-barrels

14. What are zooplankton?

2

Zooplanktons are microscopic animals living near to the surface of the water body. They are poor swimmers, They feed upon phytoplanktons, bacterioplanktons, or detritus (i.e. marine snow)

15. Write the applications of cloning vector.

2

Method of gene **cloning** is useful in studying the structure and function of genes in detail. Medical **Applications:** In medicine, **cloned** bacteria plays important role for the synthesis of vitamins, hormones and antibiotics. Agricultural **Applications:** **cloning** in Bacteria facilitates nitrogen fixation in plants.

16. What are xenobiotic?

2

XENOBIOTICS' are the foreign chemical substances found within an organism, which are not normally naturally produced by an organism or expected to be present within that organism.

17. Ex-situ technique?

2

ex situ methods is that the soil is removed from where it originally lay, for treatment.

18. What is Glyphosate

2

'Glyphosate' (herbicide) is an analogue of phosphoenol pyruvate and inhibits the enzyme 5-enolpyruvylshikimate-3-phosphate synthase. The gene coding for this enzyme has been identified, isolated and inserted into a number of plants e.g., Petunias. The gene was expressed behind a CaMV promoter and introduced using *A. tumefaciens*

19. What is cynophyta?

2

Cynophyta, a type of phytoplankton, is one particularly powerful bioindicator which is known to indicate rapid eutrophication of water bodies such as reservoirs, lakes, etc. via the creation of bloom formations.

20. What are endocrine disruptors?

2

There are chemicals which resist degradation. Partly due to lack of organisms with degradative abilities, or worse due to microbial activities that change them to more toxic form.

21. Define Bioscrubber

2

The bio scrubber is not itself truly a biological treatment system, but rather a highly efficient method of removing odour components by dissolving them. Most appropriate for hydrophilic compounds like acetone or methanol.

22. What is Sustainable bioremediation

2

23. Why DDT was banned?

2

DDT causes weakening of egg shells of birds, which led to dramatic decreases in the populations of many species. Leading to its ban. Moreover, the experience with DDT was one of the first examples that target organisms develop a resistance to the toxic chemicals. Because DDT-resistant strains of the *Anopheles* mosquito risen, DDT's value for malaria control greatly diminished

24. Define catabolic expansion. 2

25. Write two methods of Detergents bioremedies. 2

Biodegradation and

26. Consequences of OECD 2001 about global Biotechnology? 2

The OECD (2001) concluded that the industrial use of biotechnology commonly leads to increasingly environmentally harmonious processes and additionally results in lowered operating and/or capital costs.

27. Define phytodegradation and name its forms? 2

Biological breakdown of contaminants by plants.

Forms

- Internally, *having first been taken up by the plants,*
- Externally, *using enzymes secreted by them.*

28. What are living conditions of extremophiles? give examples 2

Three examples of extremophiles are *Picrophilus torridus* (a thermoacidophile adapted to hot, acidic conditions), Antarctic krill (a psychrophile), and the Pompeii worm (a thermophile).

29. Difference between bioenhancement and bioaugmentation. 2

- Bioenhancement concentrates solely on the existing micro-fauna, stimulating their activity by the manipulation of local environmental conditions.
- Bioaugmentation requires the deliberate introduction of selected microbes to bring about the required clean-up.

30. Fermentation produce a solution of ethanol in water list two method by which ethanol can be treated to

produce fixed grade ethanol 2

by subsequent simple distillation, to 95%

ethanol, or to the anhydrous form by azeotropic co-distillation using a solvent.

31. Difference between natural and conventional sludge dewatering process. 3

Natural sludge dewatering processes:

- Drying beds
- Sludge lagoons

Mechanised sludge dewatering processes:

- Centrifuges
- Vacuum filters
- Belt presses
- Filter presses

32. Enlist three attributes of aqueous invertebrate that make them ideal for use as bioindicator. 3

Invertebrates can also be Bioindicator;

- aquatic invertebrates tend to be bottom-feeders (also known as Benthos or macro invertebrates).
- Powerful indicators of watershed health as they are not difficult to distinguish in a lab,

- frequently live for more than one year,
- have restricted mobility,
- and are integrators of ecological condition

33. Difference between physical and mechanical sludge dewatering process. 3

34. Name enzyme, solution and equipment's used in the genetic engineering. 3

Restriction endonuclease or restriction enzymes, DNA ligase, DNA polymerases and reverse transcriptase

Equipment's:

Gel Electrophoresis, plasmids, transformation or transduction, PCR, Identifying transgenic organisms

35. There classes of biosensor and two advantages and disadvantages of immunsensor. 3

According to the biorecognition principle, biosensors classified into immunochemical, enzymatic, nonenzymatic receptor, whole-cell, and DNA biosensors.

Immunosensors present the advantages of sensitivity and selectivity. Limitations are the regeneration of the immunosurface, and cross-reactivity. Enzymes act as recognition elements because of their specificity.

36. Enlist the types of solid waste. (catagories likhni hain) 3

37. What are three phases present in aerated biofilter 3

Solid phase: consists of a support medium and biofilms,

Liquid phase: consists of the liquid in permanent flow through the porous medium,

Gas phase: formed by artificial aeration and by the gaseous by-products of the biological activity.

38. Phytoplankton? 3

Phytoplankton are microscopic marine algae. **Phytoplankton** is the base of several aquatic food webs. In a balanced ecosystem, they provide food for a wide range of sea creatures including whales, shrimp, snails, and jellyfish.

39. How quality of crops should enhance? 3

1. Plant Early, Plant Effectively. ...
2. Practice Seasonal Soil Rotation. ...
3. Know The Yield Potential. ...
4. Always Scout Your Fields. ...
5. Ensure Proper Water Drainage. ...
6. Utilize Fertilizers. ...
7. Test Your Soil. ...
8. Weed Early and Often.

40. Write 3 oxidation products of TNT. 3

Reduction products:

- 4-amino-2,6-dinitrotoluene (4ADNT)m
- 2-amino-4,6-dinitrotoluene (2ADNT).
- 2,4-diamino-6-nitrotoluene (2,4DANT),
- 2,6-diamino-4-nitrotoluene (2,6DANT).

Oxidation products: nitrobenzene, benzaldehydes, nitrophenol and azodicarboxylic acids

41. Process of thickening of sludge.

3

It is a physical process of concentrating the sludge, with the aim of reducing its water content and, as a result, its volume, facilitating the subsequent sludge treatment stages.

The main processes used for sludge thickening are:

- gravity thickeners
- dissolved air flotation
- centrifuges
- belt presses

42. Write features of phenols that make it pollutant?

3

- Highly toxic
- Possible accumulation in environment
- Mimic the activity of estrogen

43. What is T-DNA?

3

The transfer DNA is the transferred DNA of the tumor-inducing plasmid of some species of bacteria such as *Agrobacterium tumefaciens* and *Agrobacterium rhizogenes*. The T-DNA is transferred from bacterium into the host plant's nuclear DNA genome.

44. What is Environmental biotechnology

3

Environmental biotechnology is the branch of **biotechnology** that addresses **environmental** problems, such as the removal of pollution, renewable energy generation or biomass production, by exploiting biological processes

45. methods to get a good quality crop

3

1. Plant Early, Plant Effectively. ...
2. Practice Seasonal Soil Rotation. ...
3. Know The Yield Potential. ...
4. Always Scout Your Fields. ...
5. Ensure Proper Water Drainage. ...
6. Utilize Fertilizers. ...
7. Test Your Soil. ...
8. Weed Early and Often.

46. biological remediation.

3

Biological:

This involve the transformation or mineralisation of contaminants to less toxic and more mobile, forms.

This can include fixation or accumulation in harvestable biomass crops.

47. Write the fundamental requirements for all cloning processes.

3

Fundamental requirements of all cloning procedures:

the enzymes, solutions and equipment necessary to perform the procedures; the desired piece of DNA to be transferred; a cloning vector; and the recipient cell.

It is also essential to have some means of determining whether or not the transfer has been successful. This is achieved by the use of marker genes.

48. Advantages of association of fungi with plants

3

- Aid the growth of the plants as a result of their mycelia reaching far out into the surrounding soil
- assisting the plant in nutrient uptake

49. Enlist any three main focus areas in the global environmental market

3

Major areas so far:

- energy production,
- waste management,
- land remediation,
- water treatment.

50. With help of named example discuss how whole organisms can be used to measure potential biological

Toxicity of water soil sample?

3

Toxicity assays Microtox® (Azure, Bucks, UK), or ToxAlert® (Merck, Darmstadt, Germany). Based on the use of luminescent bacteria, *Vibrio fischeri*, to measure toxicity from samples.

51. Describe importance of anaerobic zone in biological process of phosphorus removal in activated sludge treatment?

3

The anaerobic zone gives good conditions for the development or selection of a large population of phosphorus accumulating organisms.

52. Describe natural process to remove pathogenic environment.

5

Ans: Natural:

Maturation Ponds, Land treatment.

Maturation ponds:

Shallow ponds, where the penetration of solar UV radiation and unfavorable environmental conditions causes a high mortality of the pathogens.

The maturation ponds do not need chemical products or energy, but require large areas. They are highly recommended systems...

Land treatment (infiltration in soil):

The unfavorable environmental conditions in the soil favor the mortality of the pathogens. Chemical products are not needed. Requires large areas.

53. How xenobiotic can be degraded by microorganisms

5

Ans: **Gratuitous degradation:** a xenobiotic resembles a natural compound so that it is recognized by the organism's enzymes and may be used as food, OR **Cometabolism:** a xenobiotic is degraded by the organism's enzymes but in this case its catabolism does not provide energy and so cannot be the sole carbon source. The ability of a single compound to be degraded can be affected by the presence of other contaminants. This being the case, model studies predicting the rate of contaminant degradation may be skewed in the field. Soil microorganisms are very versatile and can adapt to a food source due to exchange of plasmids.

In bacteria, the genes coding for degradative enzymes are often arranged in operons, which usually are carried on a plasmid.

Fast transfer from one bacterium to another.

Latent pathways in many Archaea.

Genetic modifications in laboratories.

54. Environmental concern to the release of the hormone into environment.

5

Ans: Endogenous hormones, of human or animal origin, have been reaching the environment for thousands of years... Exogenous steroids, used as growth promoters in several countries, have also

become a matter of concern. Endocrine-disrupting activity in aquatic fauna or even terrestrial. Their widespread use and their capability to induce responses in fish at concentrations as low as ng/l or even pg/l level, have alerted scientists to the potential dangerous consequences. An optical immunosensor to determine estrone, along with other organic pollutants (atrazine and isoproturon). New biosensing strategies for the control of hormone residues in an effort to improve food quality controls and to protect public health.

55. Define Vitrification and describe the biological process of soil remediation 5

Ans: Solidification / vitrification:

- It is the encapsulation of contaminants within a monolithic solid of high structural integrity, with or without associated chemical fixation.

Biological process of soil remediation :

This involve the transformation or mineralisation of contaminants to less toxic and more mobile,forms. This can include fixation or accumulation in harvestable biomass crops.

Advantages:

can destroy a wide range of organic compounds,can cause benefit to soil structure and fertility,are non-toxic.

Disadvantages:

the process end-point can be uncertain and difficult to gauge, treatment can be slow not all contaminants are conducive to treatment by biological means.

56. Role of filter medium in biofilter also name the compound of the filter medium. 5

Ans: The medium offers good water holding capacity and soluble chemicals within the waste gas dissolve into the film of moisture around the matrix. Bacteria, and other micro-organisms present, degrade components of the resultant solution, thereby bringing about the desired effect. The medium itself provides physical support for microbial growth, with a large surface area to volume ratio, high in internal void spaces and rich in nutrients to stimulate and sustain bacterial activity. Biofilters need to be watered sufficiently to maintain optimum internal conditions, but waterlogging is to be avoided as this leads to compaction, and hence, reduced efficiency. Properly maintained, biofilters can reduce odour release by 95% or more.

57. Five impact of chlorobenzene & dioxin on human health. 5

Ans: **Dioxin impact**

chloracne, weight loss, insomnia, liver, dysfunction, spontaneous abortion, and possibly cancer

chlorobenzene Impact

Unconsciousness, tremors, restlessness, brain, kidney and liver cancer

58. write a note on cDNA libraries 5

Ans: cDNA libraries

In eukaryotes, the first product of transcription from DNA is not messenger RNA (mRNA) but heterogeneous nuclear RNA (hnRNA). This is mRNA prior to the removal of all the non-coding sections, or introns, which are discarded during the processing to produce the mature mRNA. cDNA

is DNA which has been artificially made using the mature mRNA as a template (through Reverse Transcriptase enzyme). It is then used as the template for the second strand. Thus the synthetic DNA product is simply a DNA version of the mRNA and so should overcome the problems of expression mentioned ago.

59. write the advantages and disadvantages of incineration.

5

Ans: Advantages:

- Drastic volume reduction,
- Sterilization.

Disadvantages:

- High costs,
- Ash disposal,
- Atmospheric pollution.

60. Write the five properties of soil.

5

Ans: In case of soil particularly, properties such as depth of soil, texture, type, porosity, humus content, moisture, microbial complement, and biological activity can all have a effect on outcome of a pollutant.

61. Write five advantages of Biofilter

5

Ans:

- holds a filter medium of organic material such as peat, heather, bark chips and the like
- biofilter can reduce odour release by 95% or more.
- maintain optimum internal conditions
- The gas to be treated is forced, or drawn, through the filter
- offers good water holding capacity and soluble chemicals within the waste gas

62. Write the main objective of sludge dewatering.

5

Ans: The main objectives:

- reduction of the transportation costs;
- improvement of the handling conditions;
- increase of the calorific value of the sludge (for incineration);
- reduction in the volume for disposal;
- reduction in the production of leachate when the sludge is disposed of in landfills

63. Write five advantages of Biofuels.

5

64. List any five properties that are exploited by biosensors based on optical transducer.

5

65. Explain Vitrification methods

5

66. Why DDT banned reason of its banned although it is useful method for removal of insects?

Ans:

DDT causes weakening of egg shells of birds, which led to dramatic decreases in the populations of many species. Leading to its ban. Moreover, the experience with DDT was one of the first examples that target organisms develop a resistance to the toxic chemicals. Because DDT-resistant strains of the *Anopheles* mosquito risen, DDT's value for malaria control greatly diminished

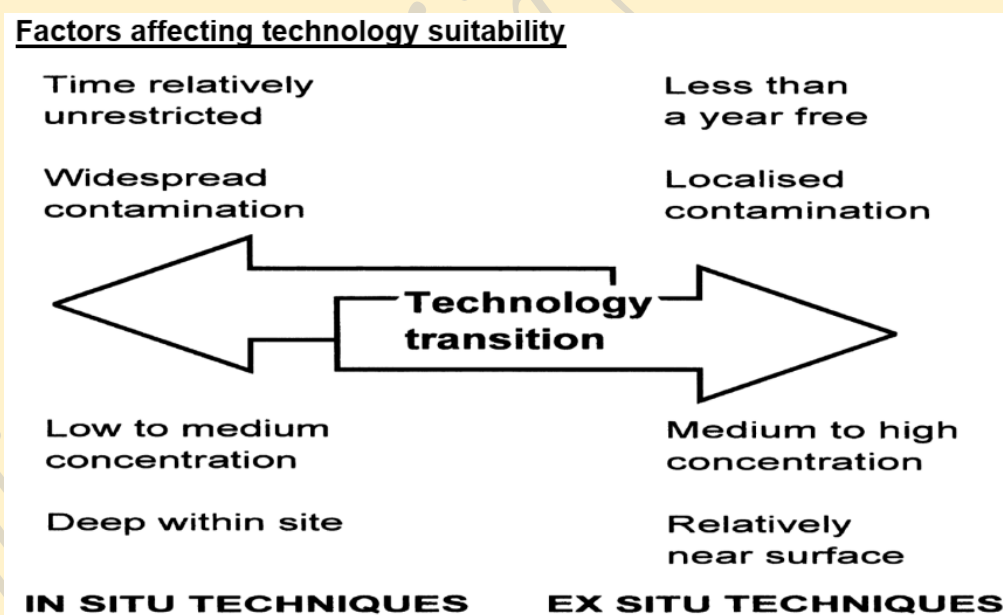
67. Genetic modification a popular plant is used for removal of mercury. describe how? 5

Ans: Genetic modification of a poplar to enable mercury to be removed from the soil and converted to a form able to be released to the atmosphere. mer A gene is one of a cluster of genes involved in bacterial detoxification of mercury, and is the one coding for the enzyme, mercuric ion reductase, which converts mercury from an ionic to a volatile form.

Initially the gene was transferred to *Arabidopsis thaliana*. The gene was transferred by gene guns to poplar tree (*Liriodendron tulipifera*) embryogenic material. The resulting poplar plantlets were found to exhibit tolerance to mercury and to volatilise it at 10 times the rate observed in untransformed yellow poplar plantlets.

68. Factors affecting the techniques of Insitu and exsitu. 5

Ans:



69. What are applications of biosensors in detection of biocides? (PPT230) 5

70. Explain role of enzyme, solution & equipment in genetic engineering? 5

Ans: the enzymes, solutions and equipment necessary to perform the procedures; the desired piece of DNA to be transferred; a cloning vector; and the recipient cell.

It is also essential to have some means of determining whether or not the transfer has been successful. This is achieved by the use of marker genes.

71. Explain genomic libraries? 5

Ans: **Genomic libraries**

Collection of total Genomic DNA from an organisms. The genomic DNA is isolated, purified and cut up into pieces of a size suitable to be inserted into a cloning vector, following by ligation and transformation into a host cell.

These DNA pieces may either be ligated as a total mixture, into a suitable vector to produce a genomic library, or a specific piece may be isolated and prepared as described ago. Genomic libraries are very useful, as they may be amplified, and accessed almost limitlessly, to look for a specific DNA Sequence. If the genomic library is of a eukaryotic origin the genes will contain noncoding regions, called introns. (Prokaryotes do not contain introns). This is a problem if the gene is to be expressed. This problem can be avoided by using cDNA.

72. What are environment factor that effect bioremediation briefly describes?

5

Ans: **Environmental Factors:**

Most significant are:

Temperature...

- 0 and 50 °C. Most efficient: 20–30 °C

pH...

- pH range of 5.0-9.0, however, 6.0-8.0 being most efficient.

Soil type...

- sands and gravels are the most suitable soil types for bioremediation...

Other factors:

Nutrient availability, oxygenation, presence of other inhibitory contaminants whether the site is contained or if the groundwater runs off, what contaminants are present, what is their concentration and whether they are biodegradable, etc. The whole issue may be viewed as hierarchical. The primary influence consists of the contaminants themselves and actual origin

73. List any five source of solid waste?

5

Ans: **Sources of solid waste:**

- Medical centers
- Food stores
- Feeding centers
- Food distribution points
- Slaughter areas
- Warehouses
- Markets
- Domestic areas

74. Mention some metallic coagulant for sludge dewatering?

5

Ans: The most common metallic coagulants are:

- aluminium sulphate
- ferric chloride
- ferrous sulphate
- ferric sulphate
- quicklime/ hydrated lime

75. Transforming problems

5

76. Write a brief note on PCBs and the use of biosensors / What are PCBs? Which biosensors detect them? 5

Ans: Polychlorinated biphenyls (PCBs) are ubiquitous environmental pollutants. Widely used as industrial chemicals, particularly as dielectric fluids in electrical transformers and capacitors. Though the production of PCBs has been banned, these are still present in the environment. The high toxicity represents a risk for public health. Different biosensor configurations to determine PCBs in the environment. DNA biosensor with chronopotentiometric detection, and various immunosensors with fluorescence, SPR, and electrochemical detection principles. Dioxins released as by-products in a number of chemical processes. Production of some pesticides, the manufacture of PVC plastics, the chlorine bleaching of pulp and paper, and waste incineration.

Carcinogenic.

A significant number of immunoassays for dioxins. Biosensors developed for PCBs have been used for Dioxins too. Biosensor for detection of dioxin-like chemicals (polyhalogenated dioxins, furans, and biphenyls) based on a recombinant mouse hepatoma cell line.

77. Phytovolatilisation

10

Ans: Phytovolatilisation.

Involves the uptake of the contaminants by plants and their release into the atmosphere, typically in a modified form. Relies on the transpiration pull of fast growing trees, which accelerates the uptake of the pollutants in ground water solution, which are then transformed and released through the leaves. E.g., genetically modified variety of the Yellow Poplar, *Liriodendron tulipifera* Engineered by the introduction of mercuric reductase gene (mer A). This confers the ability to tolerate higher mercury concentrations, and removal of the pollutant from the soil and volatilize it. In a number of studies, poplars have been shown to be able to volatilize around 90% of the TCE (Trichloroethylene) they take up

78. Describe all the sludge treatment stages before final disposal steps. (PPT_123)

10

79. Define bioremediation and explain its three routes

10

80. Giving relevant example, describe how plants may be used as bioindicator to provide information about environmental health in land.

10

81. Three classes of biosensors signal transduction and biorecognition principle.

10

Ans: Classified according to the signal transduction and the biorecognition principles. On the basis of the transducing element, categorized as electrochemical, optical, piezoelectric, and thermal sensors.

Electrochemical biosensors, and among them the amperometric and the potentiometric ones, are the best described in the literature. Those based on optical principles are the next most commonly used transducers. Various types of optical transducers exploit properties such as Simple light absorption,

Fluorescence / phosphorescence, Bio/ chemiluminescence, Reflectance, Raman scattering, and Refractive index. Surface Plasmon resonance (SPR) with main advantage that the analyte presence can be determined directly, without the use of labeled molecules. Cantilever biosensors are based on the bending of silicon cantilevers caused by the adsorption of target molecules onto the cantilever surface, where receptor molecules are immobilized. According to the biorecognition principle, biosensors classified into immunochemical, enzymatic, nonenzymatic receptor, whole-cell, and DNA biosensors. Immunosensors present the advantages of sensitivity and selectivity. Limitations are the regeneration of the immunosurface, and cross-reactivity. Enzymes act as recognition elements because of their specificity. In general, enzymatic biosensors are based on the selective inhibition of specific enzymes by different classes of compounds. Biosensors based on natural receptors can be built by *integrating the specific receptor within a membrane, and by coupling it to a transducing device*. The binding signal is detected as a structural change or an associated enzyme activity. Whole cells of living organisms *bacteria, yeast, fungi, plant and animal cells, or even tissue slices* Used as the recognition component by interrogating their general metabolic status. Useful to determine whether a substance is toxic to certain cells. In the case of DNA biosensors, *two strategies*: hybridization detection of nucleic acid sequences from infectious microorganisms, and monitoring of small pollutants interacting with the immobilized DNA layer (drugs, mutagenic pollutants, etc.) A step in the development of biosensors is the immobilization of the biological component at the transducer surface. Procures both the stabilization and the proximity between the biomaterial and the transducer.

Immobilization methods mostly employed :

physical adsorption at a solid surface, cross-linking between molecules, covalent binding to a surface, entrapment within a membrane, surfactant matrix, polymer or microcapsule...

82. Write the procedure of Sludge treatment

10

Ans: Sewage sludge treatment describes the processes used to manage and dispose of sewage sludge produced during sewage treatment. Sludge is mostly water with lesser amounts of solid material removed from liquid sewage. Reducing sludge volume may increase the concentration of some of these toxic chemicals in the sludge.

83. Explain western and Northern blotting.

10

Ans: **Northern Blotting**

- Northern blotting is a technique for detection of specific RNA sequences.
- Developed by James Alwine & George stark.
- RNA molecules have defined length & much shorter than genomic DNA it is not necessary to cleave RNA before electrophoresis.
- RNA is more susceptible to degradation than DNA.
- RNA sample are separated based on size by gel electrophoresis.

Western Blotting

- A technique for detecting specific proteins separated by electrophoresis by use of labeled antibodies.
- It is developed by George starks group in 1979
- Western blotting is based on the principles of immunothromatography where proteins were separated into poly acrylamide gel according to the isoelectric point and molecular weight.

- Involve SDS PAGE
- Use primary and secondary antibodies.

Southren Blotting

- It is a technique used for identifying specific sequences of DNA
- Developed by Edward M. Southren in 1975
- Involves agarose gel electrophoresis
- The key to this method is hybridization
- Used in gene mapping, gene discovery, diagnostic and forensic

84. Explain Cloning Vector.

10

Ans: A cloning vector is frequently a plasmid or a bacteriophage (bacterial virus) Must be fairly small and fully sequenced, able to replicate itself when reintroduced into a host cell, thus producing large amounts of the recombinant DNA for further manipulation. Also it must carry on it 'selector marker' genes, which are indicators of genomic integrity and activity. A common design of a cloning vector is one which carries two genes coding for antibiotic resistance. The 'foreign' gene is inserted within one of the genes so that it is no longer functional therefore it is possible to discriminate by standard microbiology techniques, which bacteria are carrying plasmids containing recombinant DNA and which are not. To make the manipulations easier, selector marker genes normally contain a multi cloning site (MCS) which is a cluster of sites for restriction enzymes.

Insertion of foreign DNA in vector at MSC site causes disruption of these marker genes resulting in loss of the function of that gene. *For example*, if that gene codes for antibiotic resistance, it will no longer be functional and will not protect the bacterium from that antibiotic.

Viruses may be used as vectors i.e., the DNA of viruses can be used to carry foreign DNA. Most such recombinant viral DNA vectors need phage particles to perform proper gene transfer i.e. transduction. Transfer of naked viral recombinant vector to host cells is called transfection, and tends to have lower uptake rates as compared to transduction.

85. Briefly describe different categories of Solid waste

10

Ans: Different categories of solid waste include:

- **Organic waste:** Waste from preparation of food, market places, etc.
- **Combustibles:** Paper, wood, dried leaves, packaging, etc. (high organic and low moisture content)
- **Non-combustibles:** Metal, tin cans, bottles, stones, etc.
- **Ashes/dust:** Residue from fires used for cooking.
- **Bulky waste:** Tree branches, tyres, etc.
- **Dead animals:** Carcasses of domestic animals and livestock.
- **Hazardous waste:** Oil, battery acid, medical waste.
- **Construction waste:** Roofing, rubble, broken concrete, etc.

86. What are risks of Solid Wastes regarding transfer of disease?

10

Ans: Associated risks: Disease transmission

Decomposing organic waste attracts animals, vermin and flies. Flies may play a major role in the

transmission of faecal-oral diseases. Rodents may increase the transmission of diseases such as Leptospirosis. Solid waste may also provide breeding sites for mosquitoes...

Aedes genus... Dengue fever,

Anopheles genus... Malaria...

In times of food scarcity, members of the affected population may be attracted to waste heaps to scavenge for food. Likely to increase the risk of gastro-enteritis, dysentery and other illnesses.

87. Explain the transformation method in plants (PPT25-26) **10**

88. Explain the initial steps of solid waste. **10**

Ans:

1. Identify the types of waste.
2. Identify the sources of waste.
3. Determine the potential health hazards from waste.
4. Determine the volume of waste generated
5. Identify safe collection method/s.
6. Identify safe transportation method/s.
7. Identify safe disposal method/s.

89. Difference between expression and gene reporter **10**

Ans: Expression Vectors

These are similar to the vectors described above but in addition have the required regulatory regions located before and after the 'foreign' gene which direct the host cell to translate the product of transcription into a protein. It is sometimes a difficult, expensive or time consuming procedure to analyze for product from the 'foreign' gene and so, in addition to the selector genes described above, there are frequently reporter genes to indicate whether or not the signals are 'switched on' allowing the 'foreign' DNA to be expressed. In expression cloning, there are many circumstances which can be difficult to predict.

Reporter Genes

There are many such genes in common use and these usually code for a trait or an enzyme. e.g., antibiotic resistance gene, or β -galactosidase, etc.

Blue/White screening.

Other reporter genes produce enzymes which can cause the emission of light such as the luciferase isolated from fireflies, or whose activity is easy and quick to assay like the bacterial β -glucuronidase (GUS) which is probably the most frequently used reporter genes in transgenic plants.

Reporter genes can only be a guide to the process of transcription and translation process. As with selector genes, the reporter genes serve no useful purpose once the cloning procedure has been successfully accomplished to produce the finished product.

90. How plant clean our environment and water body. **10**

Ans: Polymerase chain reaction (PCR)

A powerful technique which amplifies a piece of DNA of which only a very few copies are available. The piece must be flanked by DNA of somewhat known sequence. This allows a short sequence of DNA to be synthesized to bind specifically to the end of the sequence and act as a primer for the DNA Polymerase. A second primer is used for the other end to allow the second strand to be synthesized. The process is repeated by a constant cycling of denaturation of double stranded DNA at elevated temperature to approximately 95 °C, followed by cooling to approximately 55-60 °C to allow annealing of the primers, followed by extension of the 3' end of the primer by Polymerase usually at 72 °C. PCR requires the use of DNA polymerases able to withstand such treatment. *Thermus aquaticus* is a thermophilic microorganism from which polymerase has been isolated and used for this purpose (Taq-polymerase). DNA polymerases from other thermophiles are also being used.

MCQS:

1. What is not true about bio scrubbers?.....**Remove toxic metals.**
2. Following is true about imhoff tank used in anaerobic filter system?.....
.....**Can be a single-chamber tank or a two-compartment tank.**
3. Explosives of characterized by presence of _ group?.....**Nitro.**
4. The term "endophytes" is usually used to denote microorganisms?.....**Non-Pathogenic.**
5. UASB stands for?.....**Upflow anaerobic sludge blanket (UASB) reactors.**
6. Facultative ponds which receive raw sewage is also known as?**Primary ponds.**
7. The bio scrubber is not itself truly a biological treatment system, but rather a?
.....**Highly efficient method of removing odour components by dissolving them.**
8. Biodiesel is not derived from?**Algae.**
9. An aquatic plant?**Macrophyte.**
10. Important factors to be considered when evaluating and selecting unit operations and processes except?
.....**Sterilization of vessels.**
11. Dioxins released as by-products in a number of chemical processes?.....**Do not pose threats to environment.**
12. Nonspecific enzymes are?**oxygenase's.**
13. Wastewater is applied to the soil, supplying water and nutrients necessary for plant growth. Which is not true?
.....**Generation of aerosols.**
14. Which is not used regarding desulphurization of coal?**Psychrophiles.**
15. Suitable temperature is?**20—30C.**
16. Statement about phytoremediation is false?
17. What is false about Collection stage?**This is the stage when solid waste is transported to the final disposal site.**
18. False statement about on site disposal of waste?**Waste volumes are large.**
19. Which of following does not use to make vessel of biofilter?**Formic Sheet.**

20. Preliminary treatment involves the removal of?..... **Coarse Solid.**
21. In bacteria genetic transfer of materials occurs through?**Plasmid.**
22. Wolffia globosa is an important tool for?..... **Cadmium.**
23. Optimum PH for local environment conditions lies in the range?.....**6.5—7.5.**
24. _____uses High temperature for fuse contaminated material..... **Vitrification**
25. A step in the development of biosensors is?.....
.....**The immobilization of the biological component at the transducer surface.**
26. Associations of fungi with roots of vascular plants? **Mycorrhizae.**
27. The ability of a single compound to be degraded can be affected by the presence of other contaminants?
.....**Cometabolism.**
28. Which is true for Environment Biotechnology? **enhance or optimize conditions for existing biological systems to make their activities happen faster or more efficientl'ß**
29. Thermophile thrive at temperatures above?**45C.**
30. _____ are compounds which are man-made..... **Xenobiotic**
31. Natural estrogens are deactivated in humans by?**Glucuronidation.**
32. Collection of total Genomic DNA from an organism?**Genomic libraries.**
33. _____ refers to how pollutants enter a food chain..... **Bioaccumulation**
34. Which statement about Bioscrubber is FALSE?..... **They are useful for the removal of toxic metals**
35. Which of the following is NOT a possible destination of the pollutant in the soil?.....**appearance in surrounding air**
36. The presence of metals in environment may be detected visually by the use of bacteria. One such example is *Vogesella indigofera* that gives an idea of metal contamination by producing.....**Coloured pigments**
37. Which statement about chlorinated aliphatic hydrocarbons is correct?..... **oxidative dehalogenation of CAHs is carried out by facultative aerobes**
38. In the conventional activated sludge system. continuous arrival of food in the form of BOD to the aeration tank, leads to rapid and continuous bacterial growth. All of the following problems caused by increased bacterial growth. Except.....**the overall efficiency of the process is enhanced**
39. Which kind of the technique are suitable for bioremediation?..... **In-situ**
40. Unlike other marine species, _____ are characteristically untroubled by biofouling and study of their skin have established that its unique texture coupled with its inherent antimicrobial properties help keep them clean of fouling organisms.....**Sharks**
41. Which plant is used as a model to protect from various agents?.....**Tobacco plant**
42. The airflow in the submerged aerated biofilter is always _____ while the liquid flow can be _____
.....**upflow, downflow**
43. Which of the following approaches do not remove the pollutants but only immobilize the pollutants?
.....**phytostabilization**

44. The ____ product of pollutant may be more harmful than the original pollutant itself, therefore it is important to understand the chemistry of the pollutant.....**breakdown**
45. Among the first synthetic organic chemicals to create environmental problems were ____.....**Detergents**
46. Immobilization refer to the removal of contaminants, typically metals, by means of ____ by various microbes or plants species.....**adsorption**
47. Which of the following factors has no effect on the execution and implementation of environmental technology?
.....**health of the general population**
48. Dioxins have many congeners. The most toxic is ____ which is thought to be the most poisonous synthetic chemical.....**TCDD (tetrachlorodibenzo-p-dioxin)**
49. How much arsenic can be accumulated by using Pteris vittata?.....**2-3%**
50. Which of the following methods are best suited to instances of relatively localized pollution within a site, typically in 'hot-spots' of medium to relatively high concentration which are fairly near to the surface.....**Ex-Situ**
51. For land based applications, the soil type best suited biotechnological inventions are ____.....**Sand and Gravel**
52. Plants containing the novel DNA grow, a gene coding is frequently for ____.....**all of the above**
53. Deliberate transfer of a gene from one organism to another is called ____.....**Foreign gene**
54. Western blotting technique is used for the detection of ____.....**Protein**
55. The term "endophytes" is usually used to denote ____ microorganism.....**non-pathogenic**
56. Which of the following have the innate ability to degrade hydrocarbons contaminants through metabolic action and biosurfactant production.....**hydrocarbonoclastic bacteria (HCB)**
57. What is false about poor biodegradation of hydrocarbons?.....**Easily degraded by enzymatic attack**
58. Liquid biofuels are of considerable importance due to
Liquid fuels are of considerable importance due to the relative ease of transport and handing, and inherent controllability of combustion
59. Which process used in the natural process of sludge dewatering.....**Drying beds, sludge lagoons**
60. In aerobic Cometabolism of CAHs ---- is required ----
Successful Cometabolism can be achieved by ensuring that primary substrate is well in excess of that needed to overcome toxicity from the CAH or its transformation products